

Narrowneck Artificial Reef Renewal

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Abstract

The Narrowneck Artificial Reef was constructed by the City of Gold Coast in 1999-2000 as a key component of the Northern Gold Coast Beach Protection Strategy. While primarily intended to enhance coastal protection, its secondary objective was to improve surfing. It was constructed out of approximately 400 mega Geotextile Sand Containers using innovative filling and placement methods. Extensive performance monitoring and condition assessments were undertaken over subsequent years, but only minor maintenance was undertaken.

Almost 20 years later, the City of Gold Coast undertook the Narrowneck Artificial Reef Renewal. This resulted in an additional 84 Geotextile Sand Containers being added to reinstate and optimise performance while extending the service life of the structure. The design phase included consideration of two design options - the previous design shape and an amended shape intended to enhance performance against project objectives, including coastal protection, surfing amenity and safety. The “for construction” designs were tested in a physical model which allowed the final construction design to be optimised and performance expectations to be evaluated under a range of conditions.

The works were undertaken as a Design and Construct delivery model with provisional construction quantities. This allowed significant Contractor input, facilitating safe and efficient construction in what is a very challenging environment. It also allowed flexibility to consider alternate designs and to revise the construction design in response to changes in seabed levels, actual placement locations and available budget. The works including design, physical modelling, material supply and construction, were successfully delivered within a 6 month timeframe.

This paper provides a summary of the Narrowneck artificial reef renewal, including design investigations, project delivery model and ongoing monitoring.

Keywords: coastal protection, submerged, reef, Narrowneck

1. Introduction

The Narrowneck Artificial Reef (Figure 1) was constructed in 1999-2000 as a key component of the Northern Gold Coast Beach Protection Strategy. While primarily intended to enhance coastal protection along a narrow and vulnerable section of coast, its secondary objective was to improve surfing. It was constructed out of approximately 400 mega Geotextile Sand Containers (GSC) using custom plant and innovative filling and placement methods.

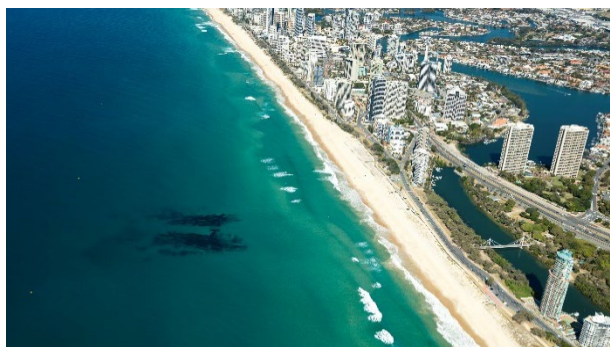


Figure 1 Narrowneck Artificial Reef (2001) on Gold Coast coastline.

The Narrowneck Artificial Reef was a prototype structure. Extensive performance monitoring (refer Section 4.1) showed that the reef has been effective in widening and increasing the resilience of Narrowneck beach in response to storm events, providing additional surfing opportunities and providing a substrate that supports a diverse marine habitat.

Condition assessments and minor maintenance were undertaken up to 2007. After almost 20 years since construction, the reef has been subject to significant seabed changes which has resulted in burial of sections of the original structure as well as the damage and subsequent loss of individual GSCs. This has resulted in a reduction of the effective reef crest level which reduced its performance as a coastal protection structure and reduced surfing amenity.

Renewal of the reef was undertaken by the City of Gold Coast (City) in 2017-2018 to reinstate and optimise performance while extending the service life of the structure.

2. Project objectives and constraints

The objectives of the Narrowneck Reef Renewal project (ICM 2017) were:

- Renewal of the reef, consistent with the original design intent and statutory approvals.
- Making use of the latest geotextile fabrics in the construction of the GSCs to ensure durability.
- Optimisation of the placement of the GSCs to work with the existing structure to promote local and global stability, maximise the service life of the structure and maintain coastal protection, with consideration of surfing amenity.
- Ensuring the works are in accordance with Safety in Design legislation considering all aspects of the asset lifecycle (design, construction, operation, maintenance and decommissioning).

The existing development approvals required the renewal design to be within the originally approved footprint, profile and construction material. The renewal works also needed to be delivered within the available budget. Constructability was also critical.

3. Delivery model

The project presented a very challenging construction environment. Given the open coast wave climate, shallow crest and variable seabed, designs needed to enable safe, practical, and efficient construction by the Contractor. Timing was also critical to allow design to be complete ready for works to be undertaken during periods when calmer conditions were expected.

A Design and Construct delivery model with a performance-based scope was adopted to facilitate successful project delivery.

The scope was not constrained by a specific reference design but focussed on project objectives and performance outcomes. An initial desktop design stage allowed for the development of two renewal options and subsequent physical modelling enabled the selection of a preferred option based on Key Performance Indicators. This delivery model allowed the design process to consider in detail relevant budget constraints and detailed input from the construction contractor to ensure that it could be successfully installed, while achieving project objectives. As the design was not known at the time of Contract Award, the construction scope provided provisional quantities of three different sizes of GSC, corresponding to sizes adopted for the original construction of the reef.

The works were undertaken by dredging contractor McQuade Marine in conjunction with design by International Coastal Management and physical

modelling by the Queensland Government Hydraulics Laboratory.

4. Renewal design

The renewal design considered the following scenarios:

- Existing reef with lower crest due to loss of crest containers – condition of reef prior to renewal works.
- Renewal Option 1 based on 2004 modified design.
- Renewal Option 2 based on 2017 modified design (developed as part of these works).

The desktop design included review and development of the concept shape as well as detailed construction designs for both renewal options (ICM 2017).

Physical modelling was undertaken of all three scenarios to establish performance under a range of conditions and allow selection of the preferred Renewal Option to be constructed (ICM 2017 and QGHL 2018).

4.1 Desktop design

As part of the desktop design development, the following was formally reviewed and summarised:

- Original design (ICM 1998)
- Formal monitoring of the structure during the period when it was regularly maintained at the design crest height (GCCM 2004, 2007 and 2011).
- Published works summarising independent monitoring undertaken, including Jackson et al. (2001, 2007, 2010 & 2012), Tomlinson et al. (2008) and Strauss et al. (2017).
- Available literature, particularly that published since the development of the 2004 modified design, including works relating to shoreline response (Ranasinghe et al. 2006 and 2010) and safety (Corbett et al. 2005 and MacMahan et al. 2011).
- Project objectives and constraints (Section 2).
- Constructability constraints for placement of GSC using nominated split-hull hopper barge.

A modified design developed in 2004 was reviewed and considered to be suitable for adoption as Renewal Option 1 (Figure 2), including:

- Crest level of -2.5m AHD (-1.5m LAT).
- Cross-shore extent of crest matches approval or previously placed GSCs.
- Alongshore reef width of 120m.
- Crest planform shape including flared shape of north and south reefs with lower bridge across 30m paddle channel (Figure 2).

The 2017 modified design was developed as part of this project for adoption as Renewal Option 2 (Figure 2), including:

- Crest level of -2.5m AHD (-1.5m LAT) as this still appropriate to achieve the project's safety objectives (Corbett et al. 2005) and construction constraints of nominated plant.
- 20m seaward shift of crest to enhance coastal protection and surfing amenity.
- Alongshore reef width of 120m.
- Crest planform generally similar in configuration with shape mildly changed, including 5° rotation of southern reef (Figure 2).

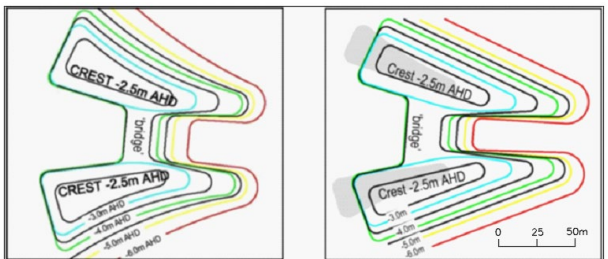


Figure 2 Concept Design for Renewal Option 1 (left) and Renewal Option 2 (right). Note Renewal Option 2 is located further seaward with slight realignment compared to Renewal Option 1 (shown in shaded grey for comparison). Source: ICM 2017

Detailed design was undertaken for both Renewal Options, including a GSC placement layout based on the pre-works multi-beam survey (Figure 3). The multi-beam survey provided an accurate representation of the existing GSC and provided a valuable tool to inform the detailed design process. This provided a high level of confidence compared with previous maintenance campaigns which were necessarily based on traditional single-beam hydrographic surveys validated using underwater site inspection and physical measurement.

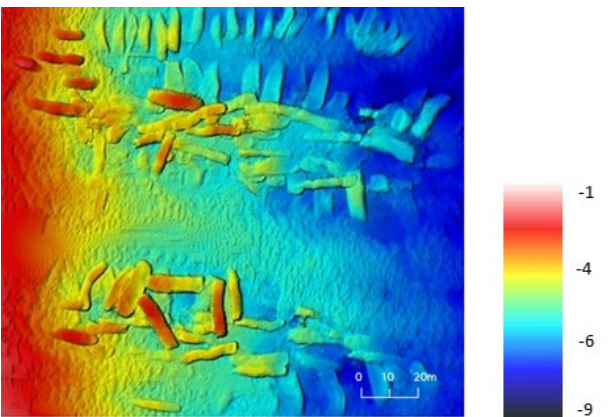


Figure 3 Pre-works multi-beam survey providing accurate representation of the existing reef, despite the complexity of the structure shape. Source: ICM 2017

The detailed design also considered the filled container geometry, flexibility of the containers when placed on a variable seabed and construction constraints including available water depth and placement orientation relative to predominant swell direction.

The detailed designs were developed considering the constraints of the available budget to achieve the concept design shape as far as possible. The north reef design shape was almost fully achieved for both renewal options, however only partial completion of the south reef and the bridge was achievable for Renewal Option 2 within the available budget.

As maintenance requirements are largely driven by individual GSC damage from anchors and knives, a more durable and robust geotextile was very desirable to reduce whole of life costs for the structure. Unfortunately, a review of commercially available products and correspondence with manufacturers did not identify any significant advances in this regard. As a result, the renewal works necessarily adopted products developed during the original construction of the Narrowneck Reef (Jackson and Corbett 2019).

4.2 Physical modelling

Physical modelling was undertaken in a 12m x 14m basin at Queensland Government Hydraulics Laboratory (QGHL 2018). A length scale of 1:43 was adopted. The general bathymetry was constructed using standard techniques; however the existing reef was Computer Numerical Controlled (CNC) router machined out of high density polystyrene foam to achieve the pre-works multi-beam survey with a 1mm model accuracy (Figure 4). The renewal designs were constructed using specially fabricated and filled GSC (Figure 5).

Preliminary model runs were undertaken as a final design tool prior to the commencement of formal physical model testing. This allowed the detailed design to be optimised through minor amendments to the container layout in response to observed performance.



Figure 4 CNC machined high density polystyrene foam replicates existing reef bathymetry with 1mm model accuracy. Individual containers are clearly apparent. Note model water level lowered to allow model setup. Source: ICM 2017



Figure 5 Renewal Option 1 constructed using scaled geotextile sand containers placed over existing reef. Source: ICM 2017

The 14 model runs for each of the 3 scenarios included a range of irregular and monochromatic wave heights between 1 - 3m and 2 different water levels. The monochromatic conditions allowed for evaluation of performance over a range of incident wave conditions and enabled direct comparison of the performance between the renewal options, particularly with regards to surfing performance and safety. Larger wave heights were not tested as coastal protection benefits were established reasonably well from monitoring of the existing structure (WRL 2019) and the larger wave heights would have required a significant reduction in the model scale within the basin, which would have made assessment of many aspects difficult.

The following Key Performance Indicators were considered:

- Frequency of wave breaking (during irregular wave conditions)
- Transmitted wave height in the lee of the reef
- Breaking wave height
- Breaker type
- Ride length
- Peel angle
- Water depth over reef during drawdown
- Wave-induced currents
- Shorebreak characteristics

4.2.1 Frequency of wave breaking

Breaking was observed on the existing reef at higher wave heights and lower tides only. As expected, both renewal options (with the crest reinstated to the design height) experienced wave breaking in a broader range of wave heights. During average wave conditions at low tide, frequency of wave breaking was increased by ~30%.

4.2.2 Breaker Type

Breaker type was classified broadly as spilling, plunging or surging. This was assessed visually during each model run, supported by oblique video recordings and photographs. Both Renewal Options experienced breaker types that were suitable for surfing (generally plunging). (ICM 2017)

4.2.3 Ride Length and Peel Angle

Peel angle and ride length was determined based on aerial video recordings (Figure 7) and generally extended from the break point probe to the intersection with the beach break (ICM 2017).

Ride lengths were similar for south reef, however for north reef (where the full design shape was able to be achieved within the budget for both Renewal Options), Renewal Option 2 (Table 4-1) was characterised by significantly longer ride lengths (35-90m compared to 35-128m, being 40% higher on average).

Table 4-1. Physical modelling results for peel angle and ride length for Renewal Option 2

Test	Tp (s)	Hs (m)	SWL	North Reef		South Reef	
				Peel angle (deg)	Ride length (m)	Peel angle (deg)	Ride length (m)
4	9.5	2.0	MSL	57	112	60	81
5	12.0	2.0	MSL	64	98	59	46
6	9.5	3.0	MSL	64	137	63	61
7	12.0	3.0	MSL	64	92	63	52
8	9.5	1.0	LAT	62	47	62	34
9	12.0	1.0	LAT	66	59	66	42
10	9.5	2.0	LAT	60	86	58	66
11	12.0	2.0	LAT	60	86	61	54
12	9.5	3.0	LAT	63	128	48	54
13	12.0	3.0	LAT	66	101	59	50

Peel angles (Figure 7) varied depending on conditions but were similar on average between both options and were within the surfable range for all tests (Table 4-1). Ride lengths and peel angles are expected to be variable in response to changes in adjacent sand bars that affect the bathymetry surrounding the reef.

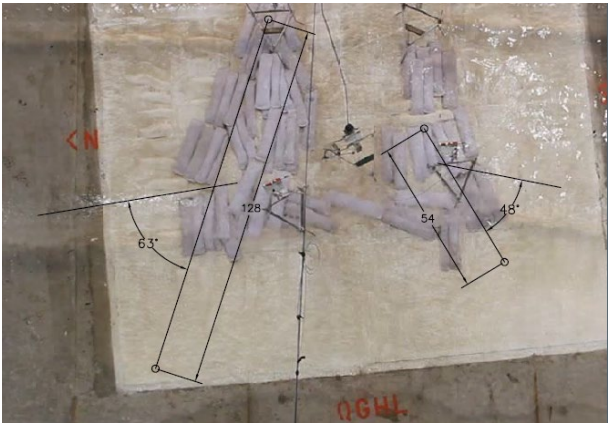


Figure 6 Annotated aerial image of physical modelling of Renewal Option 2 including measurement of ride length and peel angle. Source: ICM 2017

4.2.4 Water Depth over Reef

The point of minimum water depth over the reef was determined visually (or by physical measurement) during the calibration model run. The minimum water level required the manual relocation of individual “north drawdown” and “south drawdown”

wave probes for each model run. The minimum water depth was then determined by the difference between the level of the reef structure (scanned prior to the model runs) and the minimum water level recorded by the relevant probe.

In moderate wave conditions (circa $H_s = 1\text{--}2\text{m}$, in a typical optimal surfing range), water depths greater than 0.5m were observed. For wave heights of 3m at LAT, critical flow was observed during drawdown over containers with crest height at RL -2.5m AHD. Containers with crest levels of -2.2m AHD were observed to “suck dry” in these conditions.

4.2.5 Shorebreak Characteristics

The wave field in the lee of the reef was assessed primarily on the presence of a peeling shore break. While close-out conditions (simultaneous breaking along the wave crest with no evidence of peeling) was observed for the existing reef configuration due to uniformity of the bathymetry, peeling shore breaks were observed for both Renewal Option 1 & 2 in smaller conditions when breaking on the reef was not observed.

4.2.6 Summary

As expected, both renewal options were significantly superior to the existing reef in terms of energy dissipation, frequency of wave breaking and surfing amenity. This emphasises the importance of the crest section of the structure in achieving the performance objectives and the need for the design crest height to be achieved and maintained.

Where the theoretical design shape was largely achieved by the container placement (i.e. north reef), Renewal Option 2 provided consistently and significantly improved surfing amenity on north reef compared with Renewal Option 1, including more frequent, better quality breaks and considerably longer ride lengths (up to 128m with the break extending landward of the structure as shown in Figure 7). Where the theoretical design shape was only partially achieved (i.e. south reef), similar surfing amenity was achieved between the two renewal options.

As a result, construction of works proceeded based on the Renewal Option 2 layout.

5. Construction

Construction of the renewal works was undertaken using the methodology and plant developed for the initial reef construction. A Trailing Suction Split-Hull Hopper Dredger fills GSCs laid out in the hopper by dredging sand and pumping it directly into the GSC. The dredger is positioned with the hopper directly over the design GSC location using accurate on-board navigation systems and the GSC is placed by opening the hopper (Figure 8).

The construction was closely monitored throughout the works, with underwater visual inspection and RTK GPS staff survey of individual GSC undertaken every few days to evaluate compliance, determine water depths and final location to ensure safe and appropriate placement of adjacent GSC.

The delivery model provided flexibility that allowed rapid and effective revisions to design GSC locations in response to changes in the seabed levels due to storm events, achieved position and levels of adjacent GSC throughout construction and increases to the available budget, with consideration of both design performance and constructability.

The works, including design, physical modelling, material supply and construction, were successfully delivered in a 6 month timeframe and within budget.

The available budget and detailed design ultimately resulted in the placement of an additional 84 GSC to the reef structure.

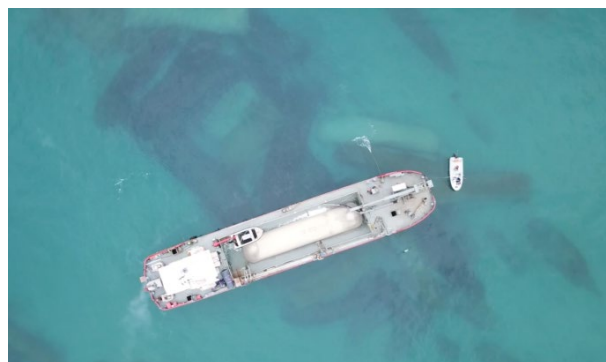


Figure 7 Trailing Suction Split-Hull Hopper Dredger at the Narrowneck Artificial Reef site prior to placing a GSC. The filled GSC can be seen in the hopper.

6. Monitoring

6.1 Coastal protection

While seasonal variability in beach width is high (approximately 40m), detailed analysis of shorelines has previously shown a salient feature about 100m to the north as well as a small groyne effect associated with the structure.

Beach widths in 2009 (when the reef was last maintained) were around 50 – 70m at Narrowneck (ETA 67 & ETA 68) (Jackson 2012). More recent analysis of beach width (September 2018 to March 2019) is showing beach widths of 70 – 95m (WRL 2019). It is noted that this is only 8 months after the renewal of the reef and as such the shoreline may not have reached equilibrium after construction. It is also possible that there may be influences from massive mega nourishment of Gold Coast Beaches although placement did not extend as far north as Narrowneck.

6.2 Surfing

The surfing amenity provided by Narrowneck Reef is provided by:

- Direct breaking of incident waves across the reef itself, typically during lower tide conditions and suitable swell height ($H_s > 1 - 2\text{m}$).
- Focusing of unbroken waves across the reef can often result in a more peaky, powerful beach break in the lee of the reef.
- Influence of the reef structure on the alignment of adjacent sand bars can often extend the reef break and/or enhance the quality of breaks in the lee of the reef and/or on the adjacent bars.

Monitoring of wave breaking on the reef through the ARGUS coastal imaging cameras (WRL 2019) indicates that (Figure 9):

- Before the reef renewal (with lower crest due to loss and/or damage of crest containers), waves would only break on the reef regularly when the RMS wave height was greater than $\sim 0.8\text{ m}$ at low tide and $\sim 1.2\text{ m}$ at high tide.
- Following the reef renewal, waves break on the reef regularly when the RMS wave height is greater than $\sim 0.5\text{ m}$ at low tide and 0.8 m at high tide.

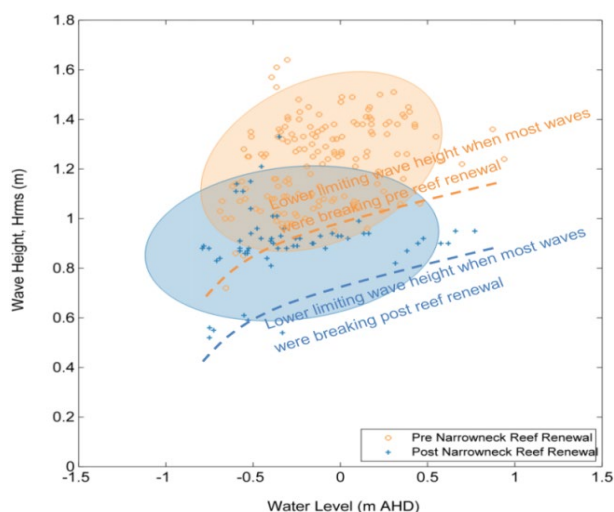


Figure 8 Comparison of wave and water level conditions where most waves were breaking on the reef before and after the reef renewal, based on coastal imaging data analysis (Source: WRL 2019)

Surfing amenity on the reef itself has been evaluated on an ad-hoc basis using surfer reporting, GPS tracking and photo and video capture. Wave breaking characteristics remain variable depending on prevailing conditions, including tide level, wave height and direction and wind speed and direction. The reef has been observed to provide high quality waves for surfing when conditions are suitable (Figure 10).

Optimal conditions on 6th July 2018 (soon after completion of the reef) resulted in a powerful and plunging break (Figure 10) with ride lengths up to 122m and 19 seconds with maximum speeds of 30.5kph recorded. Surf tracks commenced up to 62m seaward of tracks previously recorded (Figure 12). This is comparable to the results of physical modelling.

Conditions on 6th July 2018:

- $H_s = 1.6\text{m}$; $H_{\text{max}} = 3.0\text{m}$
- $T_p = 9\text{s}$
- Wave direction = 85 degrees
- Wind = 10 knots (NW) (offshore)
- Tide = 0.4m LAT

Narrowneck Reef still provides enhanced surf quality in the lee of the reef (Figure 12) and opportunities for a wide range of surf craft, including longboards, SUPs and kite boards (Jackson, 2001). More recently, hydrofoil surfing (Figure 11) has become popular at Narrowneck Reef as they are able to make use of the unbroken and re-forming waves, shoaling across the reef in conditions that are not optimal for typical shortboard surfing.



Figure 9 Surfer on Narrowneck artificial reef 6/7/2018



Figure 10 Hydrofoil Surfer at Narrowneck on 24/03/2023.

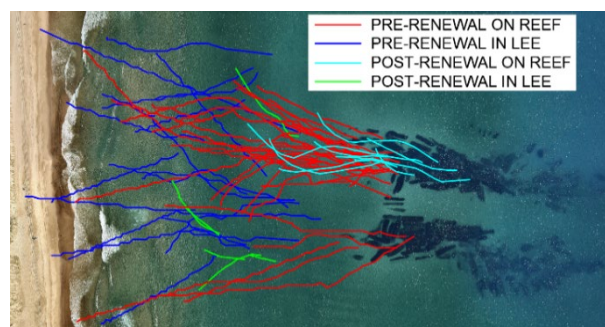


Figure 11 GPS surf tracks in the vicinity of the reef before and after renewal works. Recent surf tracks on the reef commence up to 62m further seaward than previously, enhancing surfing amenity.

6.3 Safety

The original construction of the Narrowneck Reef resulted in improved safety outcomes in the form of reduced frequency of rescues by surf life savers. In the 5 years since the Renewal was completed, the average number of rescues has remained reasonably similar. Given the variability in the data, a more detailed long-term assessment would be beneficial.

7. Summary

The Narrowneck Reef was a novel prototype design that has proven the concept of multi-functional reef structures, with the lessons learned incorporated in the Narrowneck Reef Renewal.

The Narrowneck artificial reef renewal restored the crest levels and made slight modifications to its shape to reinstate and optimise coastal protection performance, surfing amenity and safety while extending the service life of the structure.

The Design and Construct delivery model achieved an optimised performance-based design validated through comprehensive physical modelling with contractor input ensuring safe, practical, and efficient construction in what is a very challenging construction environment. It also facilitated successful delivery of the project (including design, physical modelling, material supply and placement of 84 containers) within 6 months.

Monitoring of wave breaking on the reef has shown waves break more often and in a wider range of conditions, providing more surfing opportunities. Ad-hoc monitoring of surfing amenity indicates that while the surf quality is naturally variable depending on prevailing conditions, the reef performs well when conditions are suitable, with performance consistent with physical modelling.

8. Acknowledgements

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